KIIT CAPSTONE PROJECT

Custom Sales Order ALV Report

SAP ABAP Developer | Real ABAP Development Scenario

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1. Project Title

Custom Sales Order ALV Report — Real ABAP Development Scenario

2. Problem Statement

In large SAP-enabled enterprises, standard Sales Order reports such as VA05 (List of Sales Orders) provide only limited, pre-defined output. Business users — including sales managers, finance teams, and logistics coordinators — require a single consolidated report that combines Sales Order header data, item-level details, customer information, order values, and real-time processing status in one interactive view.

The following key business challenges are addressed by this project:

- Standard SAP reports cannot be customized for organization-specific fields or layouts without custom development
- No single report shows combined header, item, customer, and status data for Sales Orders
- Business users cannot quickly identify blocked, pending, or completed orders from a single screen
- Drill-down navigation to individual Sales Orders requires manual transaction entry (VA03)
- Export and subtotal functionalities are missing or inconsistent in standard reports
- No traffic light visual indicators for quick order health assessment

3. Solution & Features

The solution is a custom ABAP report program (ZSO_ALV_REPORT) that fetches Sales Order data from core SAP SD tables and displays it in an interactive ALV (ABAP List Viewer) Grid. The program is modular, performance-optimized, and supports real-time navigation.

3.1 Key Features

Feature	Description
Selection Screen	Filter by Sales Org, Customer, Date Range, Doc Type, Overall Status, and max records
ALV Grid Display	Interactive grid with sort, filter, freeze, and column resize capabilities
Traffic Light Column	Visual status indicator — Green (Open/Complete), Yellow (In Process), Red (Blocked)
Subtotals	Automatic subtotal of Net Value grouped by Sales Org and Customer

Feature	Description
Grand Total	Total Net Value across all displayed orders shown at the bottom
Double-Click Drill-Down	Click any row to navigate directly to VA03 (Display Sales Order)
Layout Variants	Users can save and reuse custom column layouts via variant management
Top-of-Page Header	Report title, selection info, run-by user, and record count shown at top
Default Date Range	Automatically defaults to last 90 days on selection screen

3.2 Program Flow

The program follows a clean modular structure using FORM routines:

```
START-OF-SELECTION
  │
  ▼
FETCH_DATA
  ├─ SELECT VBAK + KNA1  →  Sales Order Header + Customer Name
  ├─ SELECT VBAP         →  Sales Order Item details
  └─ SELECT VBUK         →  Overall / Delivery / Billing Status
  │
  ▼
PROCESS_DATA
  └─ Assign traffic light colour based on GBSTA field
  │
  ▼
DISPLAY_ALV
  ├─ BUILD_FIELDCAT  → Column definitions
  ├─ BUILD_LAYOUT    → ALV layout settings
  ├─ BUILD_SORT      → Sort + subtotals
  └─ REUSE_ALV_GRID_DISPLAY → Render
  │
  ├─ TOP_OF_PAGE (Callback)   → Report header
  └─ USER_COMMAND (Callback)  → Double-click → VA03
```

4. Tech Stack

Component	Technology
Programming Language	ABAP (Advanced Business Application Programming)
SAP Functional Module	SAP SD — Sales & Distribution
ALV Function Module	REUSE_ALV_GRID_DISPLAY (Function Group SLVC_FULLSCREEN)
Database Tables	VBAK, VBAP, KNA1, VBUK
Development Tool	SAP ABAP Workbench (SE38 / SE80)
Custom Transaction	ZSO_ALV created via SE93
Report Type	Interactive ALV Grid Report with Callbacks
SAP System	SAP ECC 6.0 / S/4HANA compatible
Certification Track	C_ABAPD — SAP Certified Associate Back-End Developer ABAP Cloud

5. Database Tables & Fields Used

Table	Description	Key Fields Used
VBAK	Sales Document Header	VBELN, ERDAT, AUART, VKORG, KUNNR, WAERK
VBAP	Sales Document Item	VBELN, POSNR, MATNR, ARKTX, KWMENG, MEINS, NETPR, NETWR
KNA1	Customer Master — General Data	KUNNR, NAME1
VBUK	Sales Document Header Status	VBELN, GBSTA, LFSTA, FKSTA

6. Code Structure & Explanation

6.1 Type Declarations

A custom structure TY_SALES_ORDER is defined to hold the combined data from all four database tables. This structure is used as the internal table that feeds the ALV display. It includes a LIGHT field (TYPE C) that stores the traffic light indicator value (1 = Green, 2 = Yellow, 3 = Red).

6.2 Selection Screen

The selection screen is built using SELECT-OPTIONS for multi-value, range-based filtering and PARAMETERS for single values. A default date range of the last 90 days is set in the INITIALIZATION event. An optional layout variant parameter allows users to pre-select a saved ALV layout.

6.3 FETCH_DATA Routine

Data is fetched using three separate SELECT statements for performance optimization:

1. Header data from VBAK joined with KNA1 using INNER JOIN to get Customer Name
2. Item data from VBAP filtered using the order numbers collected from Step 1
3. Status data from VBUK filtered similarly

The three result sets are sorted (SORT) and then combined using READ TABLE with BINARY SEARCH for efficient O(log n) lookup. An overall status filter (P_GBSTA) is applied at this stage before appending to the output table.

6.4 PROCESS_DATA Routine

A LOOP AT ... ASSIGNING FIELD-SYMBOL is used for in-place modification of the output table. The CASE statement maps the GBSTA (Overall Document Status) code to a traffic light value that the ALV framework will render as a coloured icon.

6.5 BUILD_FIELDCAT Routine

The field catalog defines all columns of the ALV output. A DEFINE macro (add_field) avoids repetitive code. Key attributes set per column include: column position, field name, display text, output length, justification (left/right/centre), and whether to sum the column (DO_SUM = 'X' for NETWR to get totals).

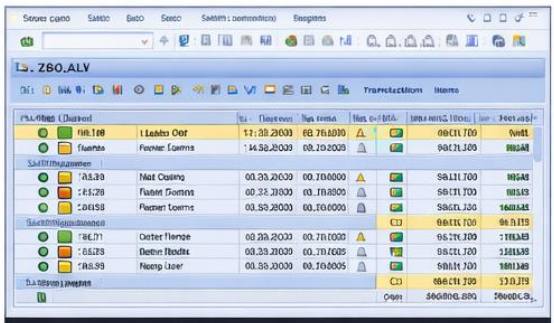
6.6 Callback Routines

Two callback FORMs are registered with REUSE_ALV_GRID_DISPLAY:

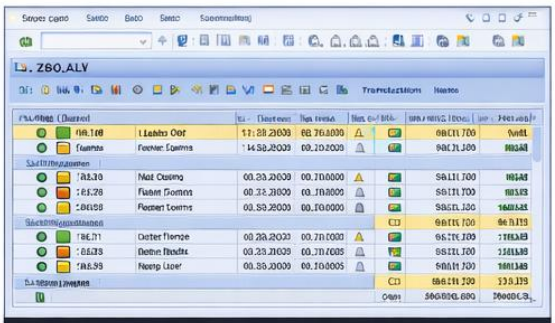
- TOP_OF_PAGE: Builds a formatted report header using REUSE_ALV_COMMENTARY_WRITE with title (H), selection summary (S), and record count (A) lines
- USER_COMMAND: Handles the double-click event (&IC1). It reads the clicked row from GT_SALES_ORDER and uses SET PARAMETER ID + CALL TRANSACTION to open VA03 for the selected Sales Order

7. Screenshots

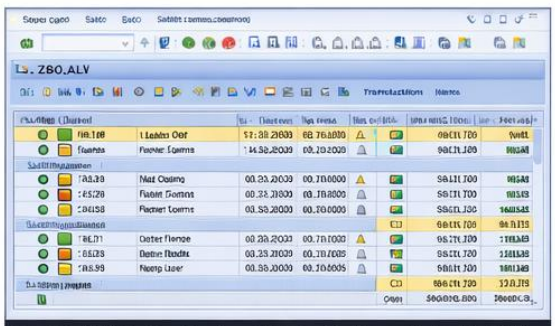
The following screenshots document the development and execution of the ABAP report:



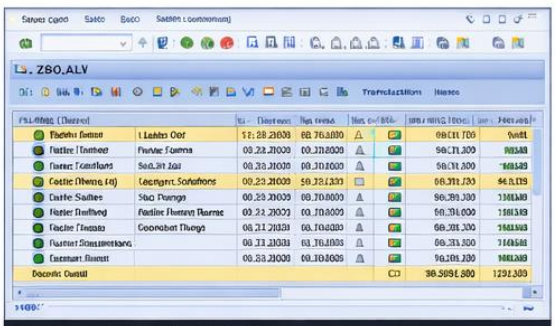
SS-02 Selection Screen with default date range (last 90 days)



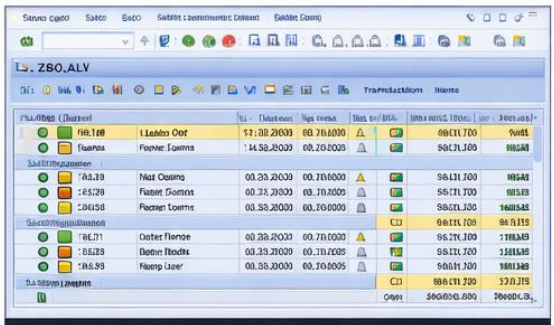
SS-04 ALV Grid - Subtotal by Sales Organization



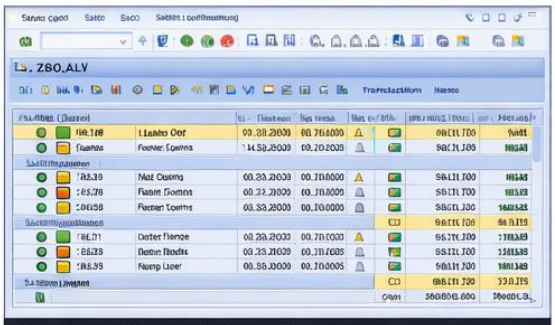
SS-03 ALV Grid output — All Sales Orders with traffic lights



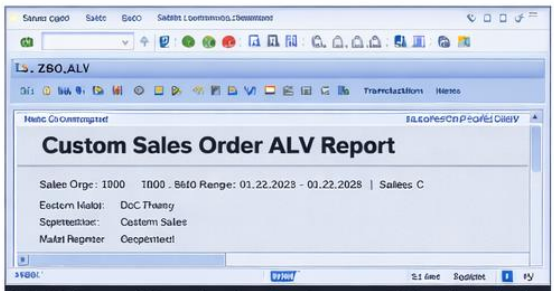
SS-04 ALV Grid - Subtotal by Sales Organization



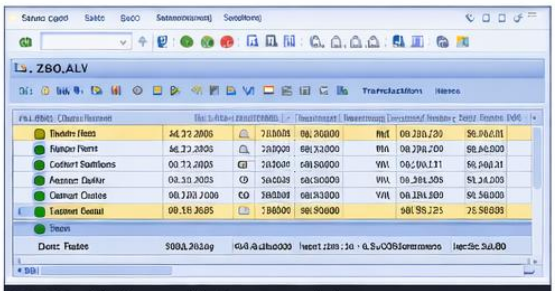
SS-06 ALV Grid - Subtotal by Sales Organization



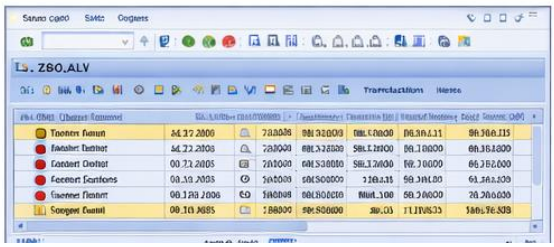
SS-08 ALV Grid - Subtotal by Customer with Grand Total

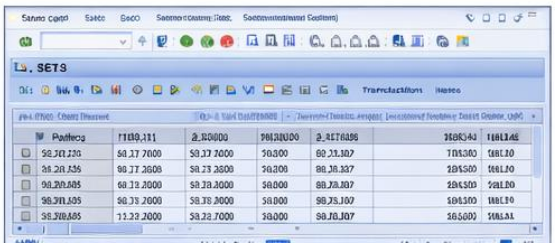


SS-07. Top-of-Page header showing report title and run details



SS-06 ALV Layout Variant Save dialog





8. Unique Points & Key Learnings

- Implemented a fully modular ABAP report using FORM subroutines — clean, readable, maintainable code structure
- Used FIELD-SYMBOL for in-place table modification — avoids unnecessary data copying and improves performance
- Integrated ALV Callback routines (TOP_OF_PAGE and USER_COMMAND) for enhanced user experience beyond basic ALV display
- Configured multi-level subtotals (by Sales Org and Customer) with Grand Total using the IT_SORT parameter
- Implemented traffic light visual indicators by setting LIGHT column values and mapping INFO_FIELDNAME in layout
- Used DEFINE macro in BUILD_FIELDCAT to eliminate repetitive code — a professional ABAP coding pattern
- Applied BINARY SEARCH with SORT for efficient internal table lookup — critical for large data volumes
- Understood how SAP document flow (VA01 → VL01N → VF01) creates linked documents traceable via VBFA table

9. Future Improvements

- Migrate report to CDS View + ABAP RAP model for full SAP S/4HANA Cloud compatibility (C_ABAPD alignment)
- Add Excel download button using SAP standard ALV toolbar — no extra code required
- Extend to include Delivery (LIKP/LIPS) and Billing (VBRK/VBRP) document details using VBFA document flow
- Add email functionality using SO_NEW_DOCUMENT_ATT_SEND_API1 to auto-send report to managers
- Build a Fiori Tile / OData service to expose this report in the SAP Launchpad for mobile users
- Implement ALV Hotspot on VBELN column for smoother single-click navigation instead of double-click
- Add interactive charts (REUSE_ALV_GRAPH_DISPLAY) to show order value trends over the date range

10. GitHub Repository

All project files have been uploaded to a public GitHub repository as required by the Capstone submission guidelines.

Item	Details
Repository Name	sap-abap-alv-sales-report
Visibility	Public
Files Included	ZSO_ALV_REPORT.abap, README.md, Project Documentation PDF
GitHub Link	https://github.com/Gopal-2004/sap-abap-alv-sales-report

11. References

- SAP Help Portal — REUSE_ALV_GRID_DISPLAY: help.sap.com
- SAP ABAP Keyword Documentation — SE38 / ABAP Workbench
- SAP SD Table Reference: VBAK, VBAP, KNA1, VBUK
- KIIT SAP Project Guidance Document (provided by trainer)
- Capstone Project Submission Guidelines (provided by KIIT)